

Laboratório 07

Letícia da Costa Venancio - 250125

Carregando pacotes

```
library(tidyverse)
library(RSQLite)
library(DBI)
```

1. Baixe o arquivo `uwmadison.sqlite3`. Conecte-se a ele usando o pacote `RSQLite`, armazenando a conexão em uma variável `conn`.

```
conn <- dbConnect(SQLite(),"database.sqlite3")
```

```
dbListTables(conn)
```

```
[1] "course_offerings"  "courses"           "grade_distributions"
[4] "instructors"       "rooms"              "schedules"
[7] "sections"          "subject_memberships" "subjects"
[10] "teachings"
```

2. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (`subjects.abbreviation='STAT'`)

Quem são:

```

SELECT DISTINCT instructors.name AS professor
FROM subjects

INNER JOIN subject_memberships
  ON subjects.code = subject_memberships.subject_code

INNER JOIN course_offerings
  ON subject_memberships.course_offering_uuid = course_offerings.uuid

INNER JOIN sections
  ON course_offerings.uuid = sections.course_offering_uuid

INNER JOIN teachings
  ON sections.uuid = teachings.section_uuid

INNER JOIN instructors
  ON teachings.instructor_id =instructors.id

WHERE subjects.abbreviation = 'STAT'
ORDER BY instructors.name;

```

Table 1: Displaying records 1 - 10

professor
ABIGAIL BENZINE
ADIN-CRISTIAN ANDREI
AKICHIKA OZEKI
ALAN HUANG
ALBERTO DEL PIA
ALBRECHT KLEMM
ALEXANDER COVINGTON
ALEXANDER FISH
ALEXANDER KISELEV
ALYSSA DIGILIO

Quantos são:

```

SELECT COUNT(DISTINCT instructors.name) AS num_professors
FROM subjects

INNER JOIN subject_memberships

```

```

ON subjects.code = subject_memberships.subject_code

INNER JOIN course_offerings
  ON subject_memberships.course_offering_uuid = course_offerings.uuid

INNER JOIN sections
  ON course_offerings.uuid = sections.course_offering_uuid

INNER JOIN teachings
  ON sections.uuid = teachings.section_uuid

INNER JOIN instructors
  ON teachings.instructor_id = instructors.id

WHERE subjects.abbreviation = 'STAT'
ORDER BY instructors.name;

```

Table 2: 1 records

num_professors
377

3. O GPA americano é definido numa escala de 0 a 4, em que A = 4, AB = 3.5, B = 3, BC = 2.5, C = 2, D = 1 e F = 0. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):

Cálculo do GPA:

- Quem é o professor mais difícil?

```

SELECT instructors.name AS professor,
  AVG((4.0 * a_count + 3.5 * ab_count + 3.0 * b_count + 2.5 * bc_count +
    2.0 * c_count + 1.0 * d_count + 0.0 * f_count) /
    (a_count + ab_count + b_count + bc_count + c_count + d_count + f_count)) AS gpa
FROM subjects

INNER JOIN subject_memberships
  ON subjects.code = subject_memberships.subject_code

```

```

INNER JOIN course_offerings
  ON subject_memberships.course_offering_uuid = course_offerings.uuid

INNER JOIN sections
  ON course_offerings.uuid = sections.course_offering_uuid

INNER JOIN teachings
  ON sections.uuid = teachings.section_uuid

INNER JOIN instructors
  ON teachings.instructor_id = instructors.id

JOIN grade_distributions
  ON sections.course_offering_uuid = grade_distributions.course_offering_uuid
  AND sections.number = grade_distributions.section_number

WHERE subjects.abbreviation = 'STAT'

AND (a_count + ab_count + b_count + bc_count + c_count + d_count + f_count) > 0

GROUP BY instructors.id, instructors.name

ORDER BY gpa ASC
LIMIT 1

```

Table 3: 1 records

professor	gpa
JAMES D KUELBS	2.6375

- Quem é o professor mais fácil?

```

SELECT instructors.name AS professor,
  AVG((4.0 * a_count + 3.5 * ab_count + 3.0 * b_count + 2.5 * bc_count +
    2.0 * c_count + 1.0 * d_count + 0.0 * f_count) /
    (a_count + ab_count + b_count + bc_count + c_count + d_count + f_count)) AS gpa
FROM subjects

INNER JOIN subject_memberships
  ON subjects.code = subject_memberships.subject_code

```

```

INNER JOIN course_offerings
  ON subject_memberships.course_offering_uuid = course_offerings.uuid

INNER JOIN sections
  ON course_offerings.uuid = sections.course_offering_uuid

INNER JOIN teachings
  ON sections.uuid = teachings.section_uuid

INNER JOIN instructors
  ON teachings.instructor_id = instructors.id

JOIN grade_distributions
  ON sections.course_offering_uuid = grade_distributions.course_offering_uuid
  AND sections.number = grade_distributions.section_number

WHERE subjects.abbreviation = 'STAT'

AND (a_count + ab_count + b_count + bc_count + c_count + d_count + f_count) > 0

GROUP BY instructors.id, instructors.name

ORDER BY gpa DESC
LIMIT 1

```

Table 4: 1 records

professor	gpa
SRIKANTHMADHAVAN ARAVAMUTHAN	4

- Qual é a disciplina mais difícil?

```

SELECT courses.name AS disciplina,
  AVG((4.0 * a_count + 3.5 * ab_count + 3.0 * b_count + 2.5 * bc_count +
    2.0 * c_count + 1.0 * d_count + 0.0 * f_count) /
    (a_count + ab_count + b_count + bc_count + c_count + d_count + f_count)) AS gpa
FROM subjects

INNER JOIN subject_memberships
  ON subjects.code = subject_memberships.subject_code

```

```

INNER JOIN course_offerings
  ON subject_memberships.course_offering_uuid =
     course_offerings.uuid

INNER JOIN courses
  ON course_offerings.course_uuid =
     courses.uuid

INNER JOIN grade_distributions
  ON course_offerings.uuid =
     grade_distributions.course_offering_uuid

WHERE subjects.abbreviation = 'STAT'

AND (a_count + ab_count + b_count + bc_count + c_count + d_count + f_count) > 0

GROUP BY courses.uuid, courses.name

ORDER BY gpa ASC
LIMIT 1

```

Table 5: 1 records

disciplina	gpa
Introduction to the Theory of Probability	2.891568

- Qual é a disciplina mais fácil?

```

SELECT courses.name AS disciplina,
  AVG((4.0 * a_count + 3.5 * ab_count + 3.0 * b_count + 2.5 * bc_count +
    2.0 * c_count + 1.0 * d_count + 0.0 * f_count) /
    (a_count + ab_count + b_count + bc_count + c_count + d_count + f_count)) AS gpa
FROM subjects

INNER JOIN subject_memberships
  ON subjects.code = subject_memberships.subject_code

INNER JOIN course_offerings
  ON subject_memberships.course_offering_uuid =
     course_offerings.uuid

```

```

INNER JOIN courses
  ON course_offerings.course_uuid =
     courses.uuid

INNER JOIN grade_distributions
  ON course_offerings.uuid =
     grade_distributions.course_offering_uuid

WHERE subjects.abbreviation = 'STAT'
AND (a_count + ab_count + b_count + bc_count + c_count + d_count + f_count) > 0
GROUP BY courses.uuid, courses.name
ORDER BY gpa DESC
LIMIT 1

```

Table 6: 1 records

disciplina	gpa
Nonparametric Statistics and Machine Learning Methods	4

4. Desconecte do banco de dados.

```
dbDisconnect(conn)
```

Horário de compilação

```
lubridate::now(tzone = "America/Sao_Paulo")
```

```
[1] "2026-09-22 20:45:46 -03"
```